

CHAPTER 1

Introduction to Computers and GUI



Learning Objectives

- What is a Computer?
- Learning Software and Hardware.
- Computer and its types.
- How does the computer operate?
- Learning Computer Operating systems.
- Learning Computer Desktop.
- Mouse Cursor Appearance.
- How to turn on and off a Computer?

"The function of good software is to make the complex appear to be simple." - Grady Booch

1. What is a Computer?

A computer is a flexible machine that follows instructions to perform various tasks, from school work to entertainment. They come in different sizes, from large ones in institutions to smaller ones like laptops and tablets at home.

2. Computer Software and Hardware

2.1 Software

Software is the computer's brain, a set of rules that guides its actions. When you play games, write, or do homework, you are using various types of software. It is the computer's secret language that brings your actions to life.

Types of Software

There are mainly two types of Software:

1. Application Software

2. System Software

Application software:

Application software is like the games and tools you use on your tablet, computer, or smartphone. These programs are made for different purposes to help you do specific things. Here are some examples of application software:

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A. Games: These are apps that you play for fun. They can be puzzle games, racing games, or adventure games.

B. Word Processors: These apps help you write and edit text. You can use them for writing stories, making school projects, or creating documents.

C. Web Browsers: A web browser is like a magic window that helps you explore the internet. You can use it to visit websites, watch videos, and find

System Software

System Software includes programs and tools that help a computer work. It is the brain and backbone of the computer. Some of the key components are:

A. Operating System (OS): An operating system (OS) is software that acts as an interface between a computer user and the computer's hardware. It manages the computer's running programs, and more. Examples include Windows, macOS, and Linux.



B. Device Drivers: Drivers are like translators that help the computer communicate with hardware like printers, mouse, and keyboards. They are needed to use a new device.

C. Utilities: Utilities are tools for tasks like file management, maintenance, and security. File managers organize files, and antivirus programs protect against malware.

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Sub-Revision

- Learn a coding language
- Learn scratch
- Interactive activities
- Step into an important field

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Windows 10

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Here's how they work together:

1. The Control Unit is responsible for directing the Arithmetic Logic Unit (ALU) to perform mathematical calculations or logical operations. This could include tasks such as addition or determining whether one number is greater than another.
2. Once the ALU completes the requested operation, it sends the result back to the Control Unit.
3. The Control Unit then stores the result in the Memory Unit and determines the next step to take.

5. Operating System

An operating system acts as the leader of a computer team, guiding it in running programs, and tasks, and communicating with other devices. The types of operating systems are as follows:

5.1 Windows

Windows is a series of operating systems developed by Microsoft. It is one of the most widely used operating systems for personal computers, laptops, and servers. Windows operating systems come in various versions, with Windows 10 being one of the latest and widely used versions.



5.2 Linux

Linux is an open-source operating system kernel. It serves as the foundation for various Linux distributions (distros) that are developed by different organizations or communities. Linux is used in servers, embedded systems, supercomputers, and as a desktop operating system.



5.3 macOS

macOS is the operating system developed by Apple Inc. It is designed exclusively for Apple hardware, such as Macintosh computers. macOS is known for its sleek and user-friendly interface, as well as its integration with other Apple devices and services.

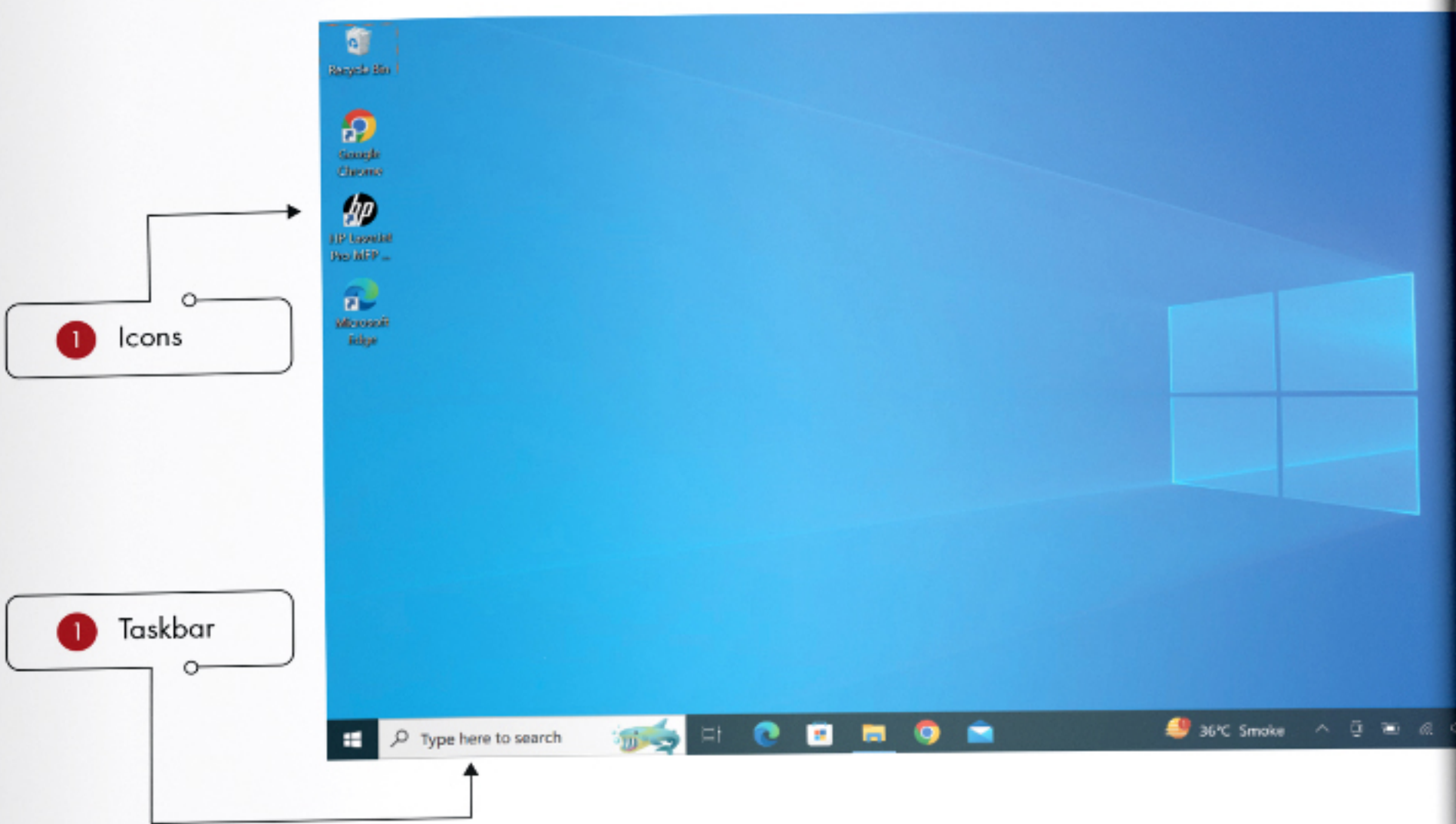


6. Windows Desktop

A desktop is the main screen of your computer, like a desk where you keep important items. The icons on the desktop are shortcuts to open programs and files.

6.1 Components of Desktop

Let's learn about each component in detail:



6.2 Icons

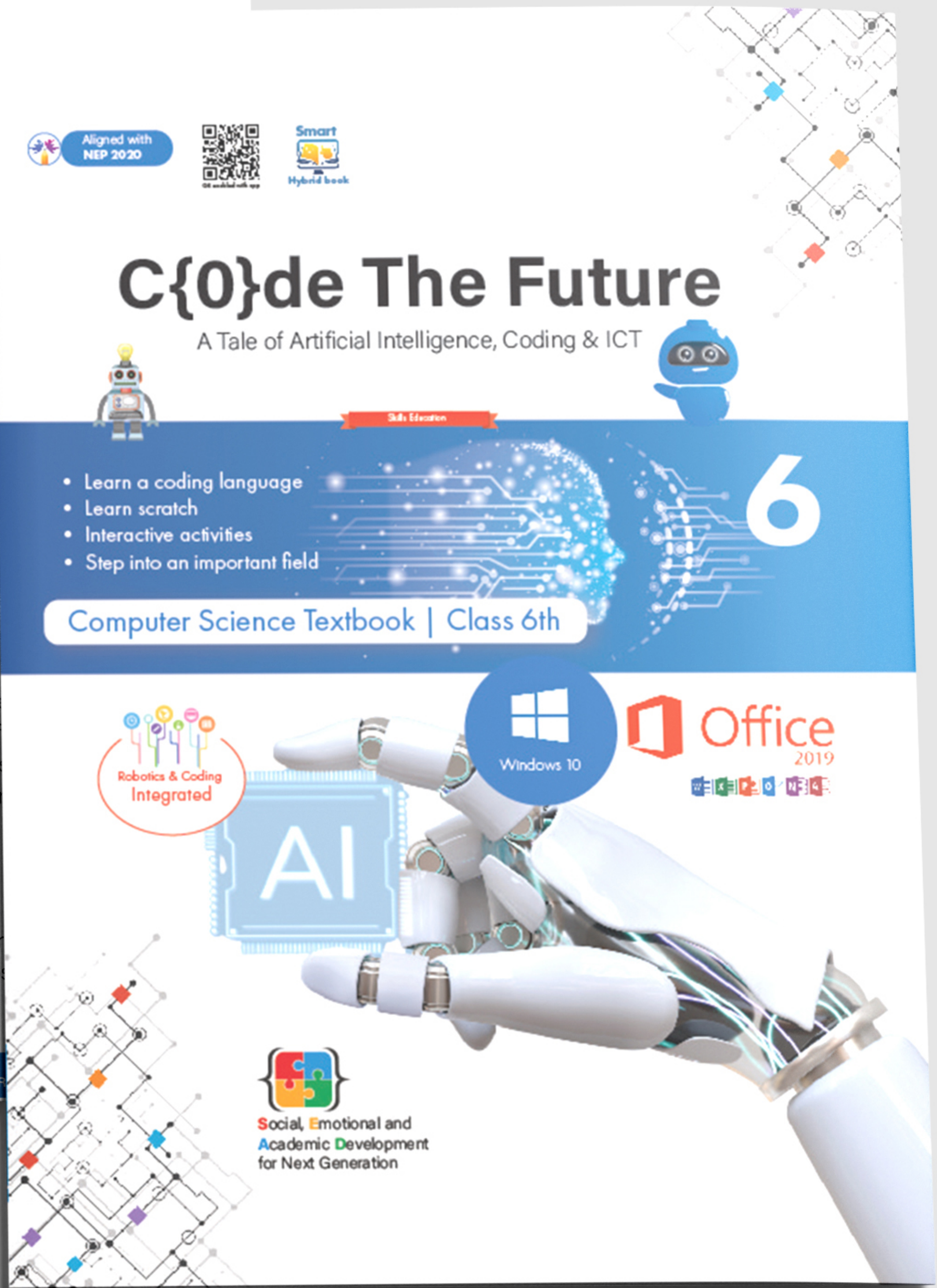
Icons are symbols that represent files or programs on a computer. They make it easy to find and open things, like pictures on a bookshelf.

6.3 Desktop Background

The desktop background, also known as the wallpaper, is the image that serves as the backdrop for icons and programs on your computer. It is a way to add a personal touch to your computer, ranging from family photos to abstract art. You can change it as often as you like, just like you would with a poster on your computer a fresh and fun look.

6.4 Taskbar

The taskbar is like a special area at the bottom of your computer screen. It holds shortcuts to programs and files, or a shelf for your computer. The taskbar does some important jobs.



New Technologies

The image appears according to the prompt



Introduction to Robotics

Robotics is like a fun world of machines and robots! Imagine having a toy that can move on its own and do cool things. Well, that's what robotics is all about. So, robotics is all about creating and using robots to make our lives easier, more fun, and sometimes even safer. It's like a big adventure in the world of technology.



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Real World Applications

Windows Laptop



1. A Windows laptop is a type of computer that runs on the Windows operating system.
2. It looks like a small, portable computer that you can carry around.
3. It's like having a mini version of a regular desktop computer, and it can do many tasks like playing games, doing homework, and browsing the internet.
4. It has a Start menu where you can find and open programs, and a variety of software and apps you can use.

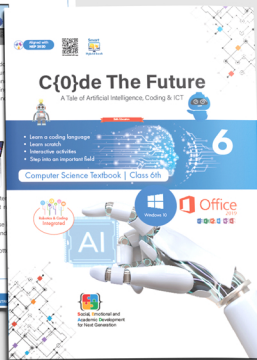
MacBooks



1. A Mac laptop, also known as a MacBook, is a type of computer that runs on the Mac OS operating system.
2. It's also a portable computer, like a Windows laptop, but it's known for its sleek design and easy-to-use interface.
3. Mac laptops are known for their sleek design and easy-to-use interface.
4. They are great for creative tasks like making art, music, and writing reports.
5. Instead of a Start menu, Mac laptops have a Dock at the bottom of the screen to access programs.

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Exercise

Fill in the Blanks

(Scanner, Speakers, Keyboard, Monitor, Waiting)

1. The _____ is like a magical talking board.
2. A computer _____ is like a TV screen for your computer.
3. _____ are computer output devices that enable you to hear sound, such as music, videos, and people talking.
4. A scanner is like a paper camera that turns drawings, photos, and words into _____.
5. The _____ cursor shows when a program is busy and can't accept input.

Write T/F for each statement

1. A webcam is a device for printing text and images.
2. Smart Watch is a type of Computer.
3. A USB flash drive is a type of storage device.
4. The desktop background is also known as the wallpaper.
5. The taskbar is located at the top of the computer screen.

MCQ's use tick (✓) for correct answer

1. Which of the following is an input device?

☐ Printer	☐ Keyboard
☐ Monitor	☐ Speaker
2. Which software manages computer tasks and runs other software applications?

☐ Application software	☐ Device drivers
☐ System software	☐ Cloud storage
3. What type of device is a USB flash drive?

☐ Input device	☐ Storage device
☐ Output device	☐ Processing device
4. Which mouse pointer shape is used for moving objects without resizing them?

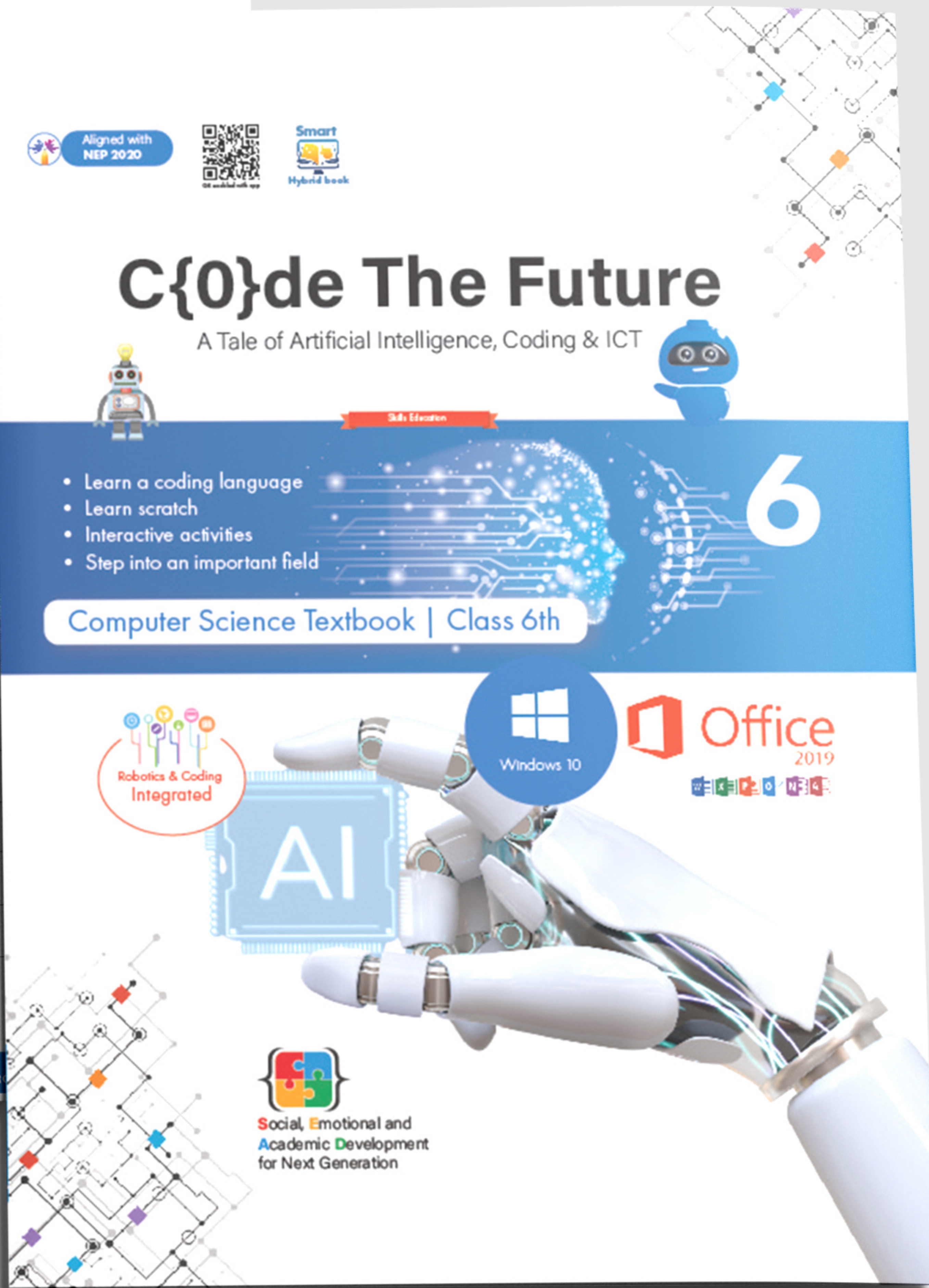
☐ Arrow Cursor	☐ Hand Cursor
☐ Text-Pointer/I-beam	☐ Move the Cursor

5. What is the primary function of the desktop background?

- | | |
|--|--|
| ☐ Protect the screen | ☐ Print documents |
| ☐ Display the current time and date | ☐ Serve as a background for program icons |

Subjective question

1. What is a CPU in a computer?
2. Describe three different types of input devices used in a computer.



Hands on

INTEGRATED ACTIVITIES SECTION

Lab session

Let's Practice

A. Look at the computer of your lab and find out different parts of the computer.
1. Write the 4 different parts of the computer you discovered or learned about.
2. Start the computer and note down your 6 favourite icons.

Group discussion

Let's Express

A. Discuss main points about GUI and how it helps us achive tasks learned in this lesson.

Project work

Let's Enjoy

A. Continue from previous Lab sessions, Download take screenshot of your favourite icons on a computer then use MS Word or Paint to print the icons and paste them below.

Teachers corner

Guided Teaching

1. Expand on the fundamental concepts of computer hardware and software.
2. Develop interactive activities or simulations that allow students to explore the graphical user interface (GUI) of various software applications and operating systems.
3. Integrate multimedia resources, such as videos, tutorials, and interactive software tools, to enhance students' understanding of computer concepts and GUI.

Learn more

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TRIVIA

Let's Practice

1. Did you know that the first-ever computer was so big it filled an entire room?
2. Your computer's desktop background can be like changing the poster on your wall. You can make it a picture you like, like your favorite superhero or a beautiful landscape.
3. The computer mouse got its name because of its shape and the way it looks like a mouse. It's not a real mouse, but it helps us control the computer!

Career corner

Business Owner

Small Business Owner:

If you have a basic understanding of computers, you can start and manage your own small business. You can use computers for various tasks, like bookkeeping and customer service.

Customer Service Representative:

Many businesses use computers to access customer information. Basic computer skills are needed to assist customers effectively.

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